

Scarlett Mukherjee

Embedded Systems & IoT Firmware Engineer • 9 Years Experience • MERIQ Index: 70% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: ARM Cortex, PCB Debugging, MQTT, Embedded Linux, I2C/SPI/UART, Python

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core ARM Cortex service, reducing latency by 34%.
- Led a squad of 8 engineers delivering scalable microservices processing 35M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 52%.

Associate Embedded Systems & IoT Firmware Engineer — *Intel* (2017 - 2023)

- Designed and maintained PCB Debugging data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$24k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [ARM Cortex, PCB Debugging, MQTT, Embedded Linux]

Engineered end-to-end production solution leveraging ARM Cortex and PCB Debugging with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [MQTT, Embedded Linux, I2C/SPI/UART, Python]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Computer Science

University of Illinois Urbana-Champaign — 2013 - 2017 (Score: 3.62 / 4.0)